

The TTLL10 polyglycyclase is stimulated by tubulin glutamylation and inhibited by polyglycylation

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In their study, Cummings et al. provide a **valuable** advance in understanding the hierarchical regulation of tubulin polyglycylation, demonstrating that TTLL8 initiates monoglycylation which is a prerequisite for TTLL10-mediated polyglycylation. The evidence supporting these mechanistic insights is **solid**, relying on a **compelling** combination of purified biochemical assays, mass spectrometry, and microscopy. The work is further valued for revealing an unexpected crosstalk between polyglycylation and polyglutamylation that ensures a balanced post-translational modification landscape for proper cilia function.

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Abstract

Microtubules in cells have complex and developmentally stereotyped posttranslational modifications that support diverse processes such as cell division, ciliary growth and axonal specification. Glycylation, the addition of glycines, singly (monoglycylation) or in chains (polyglycylation), is primarily found on axonemal microtubules where it functions in cilia maintenance and motility. It is catalyzed by three enzymes in the tubulin tyrosine ligase-like family, TTLL3, 8 and 10. We show that TTLL8 monoglycylates both α - and β -tubulin, unlike TTLL3 which prefers β -tubulin. Microscopy and mass spectrometry show that TTLL10 requires monoglycylation for high affinity microtubule binding and elongates polyglycine chains only from pre-existing glycine branches. Surprisingly, tubulin polyglycylation inhibits TTLL10 recruitment to microtubules proportional with the number of posttranslationally added glycines, suggesting an autonomous mechanism for polyglycine chain length control. In contrast, tubulin glutamylation, which developmentally precedes polyglycylation in cilia, increases TTLL10 recruitment to microtubules, suggesting a mechanism for sequential deposition of tubulin modifications on axonemes. Our work sheds light on how the tubulin code is written by establishing the substrate preference and regulation of TTLL glycyclases and provides a minimal system for generating differentially glycylation microtubules for *in vitro* analyses of the tubulin code.

Introduction

Microtubules are dynamic, non-covalent biopolymers essential for basic cellular processes in all eukaryotes. They serve as tracks for intracellular transport and build complex cellular structures such as the bipolar spindle and the axonemes in cilia and flagella. Microtubules are constructed of α/β -tubulin heterodimers which consist of a globular body, involved in polymerization interfaces, and intrinsically disordered C-terminal tails that decorate the microtubule outer surface, are heavily posttranslationally modified, and fulfill complex regulatory functions (1). Microtubule-based subcellular structures have distinct posttranslational modification patterns which regulate a microtubule's interactions with effectors and its function in cellular physiology. This is known as the tubulin code (1, 2). Tubulin modification patterns are the result of the combinatorial action of tubulin modification enzymes which write and erase the tubulin code. Aberrant patterns of tubulin modifications are associated with tumorigenesis (3, 4), poor cancer prognosis (5), neurodegeneration (6, 7), neurodevelopmental disorders (8) and other pathologies (9), and mutations in tubulin modification enzymes are linked to several human pathologies (10). It is therefore crucial to understand the substrate specificity and regulation of the enzymes that introduce these modifications.

Ciliary microtubules are among the most heavily posttranslationally modified. Glycylation, which involves the addition of glycines, singly (monoglycylation) or in chains (polyglycylation) to internal glutamates in the C-terminal tails of α - or β -tubulin, appears largely limited to cilia where it is highly abundant (11) and where it accumulates as cilia mature (12, 13). Glycylation is ATP-dependent and catalyzed by three enzymes in the tubulin tyrosine ligase-like (TTLL) family, TTLL3, 8 and 10 (14, 15). Members of this family also catalyze glutamylation (16), which is also highly abundant in cilia and flagella. Glycylation is critical for the formation, stability, and function of cilia in protists (14, 17–19), zebrafish (14) and mammals (4, 12, 20, 21). Loss of the glycyrase TTLL3 leads to a reduction in primary cilia and increased rates of cell division in colon epithelial cells (4), and TTLL3-inactivating mutations were identified in patients with colorectal cancer (4, 22). Loss of glycylation also severely impairs flagellar beating, resulting in male subfertility in mice (20). *In vitro* biophysical studies of microtubules isolated from *Tetrahymena* showed that glycylation increases microtubule stiffness (23). More recent work revealed that glycylation inhibits katanin microtubule severing (24). Thus, glycylation affects both the intrinsic biophysical properties of microtubules as well as how microtubules are recognized by effectors.

Glycylation is a polymodification. It involves two steps: initiation and elongation. During glycylation-initiation, a glycine is ligated to an internal glutamate residue through an isopeptide bond with the γ -carboxyl group. This is the branch-point glycine. In the initiation reaction, the internal glutamate residue is the acceptor amino acid, and the glycine residue is the donor amino acid (Figure 1A). In contrast, in the glycylation-elongation reaction, the glycine is ligated to the α -carboxyl group of an existing branch-point glycine or a polyglycine chain. In the elongation reaction, both the donor and the acceptor amino acids are glycines (Figure 1A). Thus, glycylation initiation and elongation necessitate the recognition of a glutamate acceptor and glycine acceptor, respectively, two amino acids with different physiochemical characteristics. Vertebrates have three glycyrases: TTLL3, 8 and 10 (14, 15). Cellular overexpression studies coupled with the use of monoglycylation-specific antibodies (15) as well as biochemical and mass spectrometry-based studies (25) established that TTLL3 is a glycylation-initiase, adding only monoglycines to internal glutamates in tubulin tails, with a preference for β -tails (25). Cellular overexpression studies coupled with the use of antibodies that recognize mono- and polyglycylation indicate that TTLL8 is also a glycylation-initiase, while TTLL10 a glycylation-elongase (15, 26). However, direct biochemical evidence with purified enzymes for segregated initiation and elongation activity for glycyrases is still lacking as does knowledge of their substrate specificity and regulation.

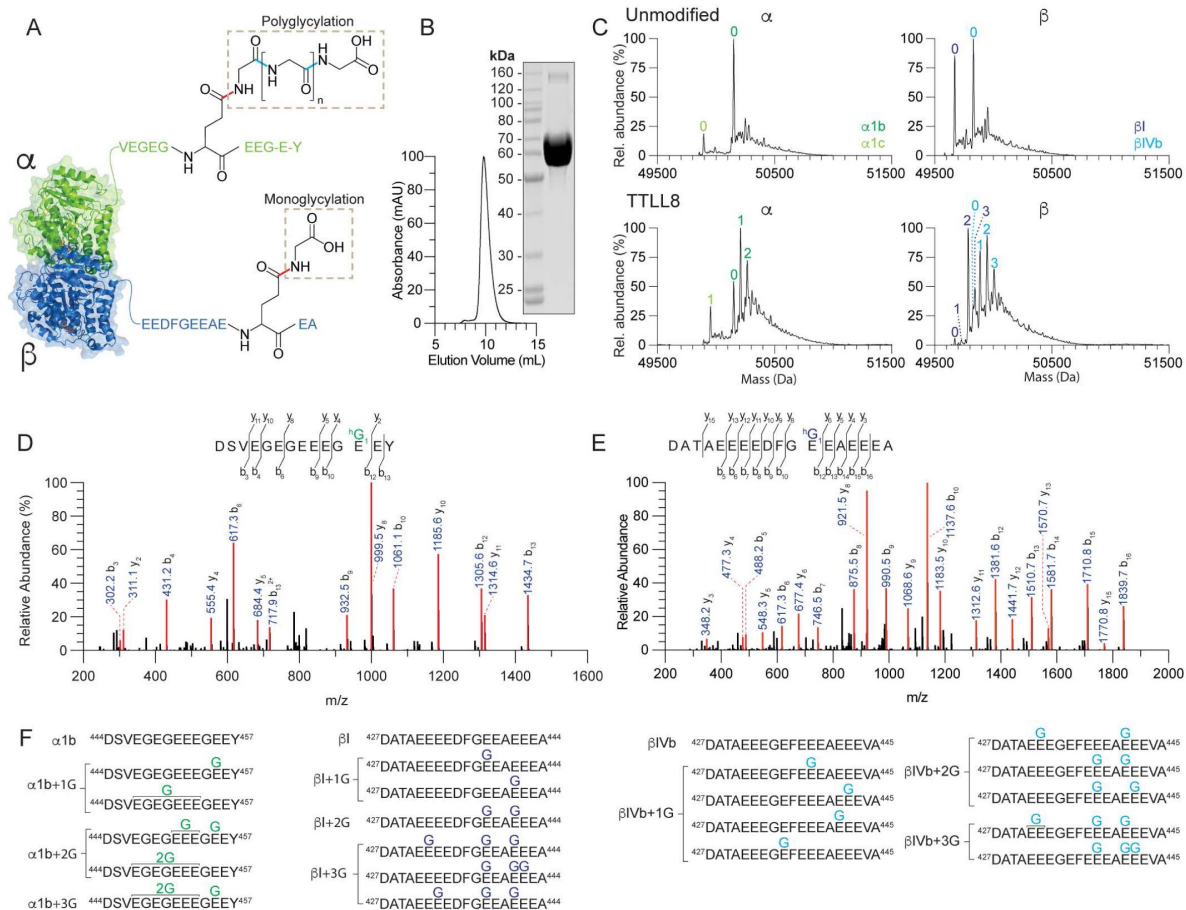


Figure 1.

TTLL8 is a monoglycylase that initiates at multiple sites on α - and β -tubulin tails.

(A) Cartoon of the $\alpha\beta$ -tubulin dimer and C-terminal tail sequences illustrating the chemical structure of monoglycine branches and polyglycine chains. In monoglycylation, the donor (incoming) glycine is linked to the γ -carboxyl group of an acceptor glutamate in the tubulin tail *via* an isopeptide bond (red). In polyglycylation, the donor glycine is linked to the α -carboxyl group of the terminal acceptor glycine in the growing glycine chain *via* a standard peptide bond (cyan). (B) Size exclusion column elution profile and Coomassie-stained SDS-PAGE gel of recombinant purified TTLL8 (C) Mass spectra of unmodified human microtubules incubated without (top) or with TTLL8 (bottom). The number of posttranslationally added glycines is indicated in green (for α -tubulin isoforms) and blue (β -tubulin isoforms). The weighted mean of the number of glycines ($\langle n^G \rangle$) added to α - and β -tubulin are denoted $\alpha + \langle n^G \rangle$; $\beta + \langle n^G \rangle$. For experimental details see Materials and Methods. (D-E) Representative MS/MS spectra from two independent experiments for monoglycylated α - (D) and β -tubulin (E) C-terminal tail peptide fragments proteolytically released from microtubules glycylation by TTLL8. For additional spectra see **Figure S2**. (F) List of TTLL8 monoglycylation sites identified by MS/MS sequencing. The line marked with 2G indicates that a monoglycine can be present at any of the glutamates in that region.

Using recombinant purified enzymes and well-defined tubulin substrates coupled with tandem mass spectrometry (MS/MS) we show that TTLL8 is exclusively a glycyl initiase, adding monoglycines to either α - or β -tubulin tails. Microscopy-based and MS/MS analyses show that TTLL10 requires monoglycylation for high-affinity microtubule binding and elongates polyglycine chains of variable lengths only from pre-existing monoglycines on both α - and β -tubulin tails. TTLL10 microtubule binding decreases monotonically with polyglycylation, indicating that TTLL10 activity is self-limiting and suggests an autonomous mechanism of polyglycine chain control. Furthermore, we find that TTLL10 microtubule recruitment is enhanced by glutamylation, a modification that precedes glycylation during cilia biogenesis (12), suggesting a potential biochemical basis for the order of deposition of these different tubulin modifications on the axoneme. Our work establishes the substrate preference and regulation of TTLL glycylation and sheds light on the biochemical mechanisms that generate tubulin modification patterns in cells.

Results and discussion

In vitro reconstitution of microtubule polyglycylation

To determine the substrate preference and biochemical activity of TTLL8, we purified recombinant human TTLL8 (Figure 1B) and assembled glycylation reactions with microtubules polymerized from unmodified human tubulin. Unmodified tubulin was isolated from tsA201 cells through a TOG affinity procedure. This tubulin consists primarily of unmodified α 1b, β I and β IV-tubulin isoforms (27). Intact liquid chromatography mass spectrometry (LC-MS) of these reactions showed that TTLL8 adds glycines to unmodified microtubules on both α - and β -tubulin (Figure 1C). Using MS/MS we mapped glycylation sites on the TTLL8-modified microtubules (Figures 1D-F, S1, S2). The MS/MS analyses show that TTLL8 adds only branch-point glycines (monoglycines) to several glutamates on both the α - and β -tubulin tails (Figures 1D, 1E, 1F, S1, S2). A tubulin with more than one branch-point glycine is still referred as monoglycylated. We could not detect any polyglycine chains, indicating that TTLL8 is an initiase, also consistent with previous reports using antibody-based assays (15). TTLL8 activity was also detectable by Western blot which showed a saturated band after treatment of tubulin with TTLL8, consistent with the mass spectrometry data (Figure S3). Extracted ion chromatographs (XIC) for α 1b peptides show a dominant peak at position E455 (Figures S1A) with additional possible modifications at positions E447-E453 (Figures 1F, S1A-C, S2A). XIC for β I peptides show a dominant peak at position E438 followed by a less abundant peak at position E441 (Figure S1D). Monoglycylation at these two sites yields the most abundant species with two monoglycines (Figure S1E). Interestingly, modification at E432 is found in the most abundant species with three or four monoglycines (Figures S1F, S1G, S2B), however modification at this position is not abundant in either the +1 or the +2 glycine peaks, indicating the lower preference of the enzyme for this site, or the possible dependence on modification at other sites prior to activity at the E432 site. Similar modification patterns were observed on β IVb (Figure S1H). Thus, TTLL8 is an initiase that adds monoglycines at multiple positions in both α - and β -tubulin tails. Our previous work with recombinant tubulin TTLL3 showed that enzyme to also be strictly a monoglycylase (25).

We then purified recombinant *Xenopus tropicalis* TTLL10 (Figure 2A) and used it in *in vitro* assays with unmodified microtubules. In contrast to TTLL8, TTLL10 glycylation only microtubules that are already monoglycylated and does not glycylation unmodified microtubules, even when incubated for a prolonged time (Figure 2B). This is consistent with cellular overexpression data which showed that polyglycylation signal was detected *via* antibody only in tubulin from cells that co-expressed TTLL8 and TTLL10, but not TTLL10 alone (15, 26). LC-MS analyses show that TTLL10 can add glycines to both α - and β -tubulin as long as they were previously monoglycylated (Figure 2B). MS/MS shows that TTLL10 is only elongating glycine chains from pre-existing monoglycines and does not initiate new glycine chains (Figures 2C-E, S4A-C, S5A-B). To

facilitate the MS/MS analyses we used heavy glycine in our modification reactions to distinguish between the TTL10 added glycines and pre-existing glycines on tubulin (Materials and Methods). XIC analysis indicates that, consistent with β I position E438 and α 1B position E455 being the dominant sites modified by TTL8, the main species in TTL10 modified microtubules contain polyglycine chains of variable lengths at α 1B position E455 and β I position E438 (**Figures S4A** [↗](#), **S4B** [↗](#), **S5A-B** [↗](#)). Thus, TTL10 elongates polyglycine chains from pre-existing monoglycines at multiple positions on α and β -tubulin tails.

TTL10 recognizes monoglycines on tubulin tails

Next, we wanted to understand the mechanistic basis for the exclusive activity of TTL10 on microtubules that already had monoglycine branches. We used total internal reflection fluorescence (TIRF) microscopy to visualize the association of Alexa647-labeled TTL10-SNAP with unmodified and TTL8-modified microtubules (Materials and Methods; **Figure 3A** [↗](#) and **Figure S6** [↗](#)). These assays show that TTL10 binds robustly to monoglycylated microtubules (**Figure 3B** [↗](#)), with an apparent K_d of ~ 600 nM, while showing little binding to unmodified microtubules. Moreover, assays with microtubules with different monoglycylation levels revealed that TTL10 recruitment to the microtubule increases with increasing numbers of monoglycines on tubulin (**Figures 3C, D** [↗](#)). Specifically, TTL10 recruitment to monoglycylated microtubules ($\langle n^G \rangle_\alpha \sim 0.8$, $\langle n^G \rangle_\beta \sim 1.2$) increases more than $\sim$ sixfold compared to unmodified microtubules. Addition of more monoglycines to the tubulin tails has a more muted effect, resulting in only a $\sim 17\%$ increase in recruitment to microtubules with $\langle n^G \rangle_\alpha \sim 1.1$, $\langle n^G \rangle_\beta \sim 2.5$ (**Figure 3D** [↗](#)). This suggests that the TTL10 active site recognizes a single branch-point glycine and that steric hindrance on the tubulin tails limits the number of branch-point glycines that can be simultaneously recognized by multiple TTL10 enzymes. The strong stimulation of TTL10 recruitment to the microtubule by monoglycylation is consistent with our activity assays that show TTL10 adds glycines only to microtubules that already have branch-point glycines on their C-terminal tails (**Figure 2** [↗](#)). This mechanism of branch-point glycine recognition contrasts with that of the glutamyl elongase TTL6 which binds to unmodified and glutamylated microtubules with similar affinity, but has a higher catalytic rate on tubulin tails that have pre-existing branch-point glutamates (**28** [↗](#)). The two-enzyme requirement for the addition of polyglycine chains to unmodified tubulin is also consistent with the chemically distinct substrates that mono- and polyglycylation requires: addition to a glutamate first (monoglycylation), followed by addition to a glycine (polyglycylation; **Figure 1A** [↗](#)) in contrast to glutamylation which uses a glutamate as a substrate for both the initiation and elongation steps.

TTL10 activity on microtubules is self-limiting

Given the pronounced effect monoglycylation has on TTL10 recruitment to microtubules, we then investigated the effect of polyglycylation on TTL10 microtubule binding. For this, we generated microtubules with different polyglycylation levels (**Figure S7** [↗](#)) and used them in TIRF-based assays. These showed that TTL10 recruitment to polyglycylated microtubules is dramatically reduced compared to monoglycylated microtubules (**Figures 4A** [↗](#), **4B** [↗](#)). Strikingly, the binding decreases monotonically with polyglycine chain length (**Figure 4B** [↗](#)). This contrasts with monoglycylation which is stimulatory for TTL10 recruitment (**Figures 3B** [↗](#), **3C** [↗](#), **3D** [↗](#)). We observed a similar inhibition of TTL10 microtubule recruitment by polyglycylation when we monitored TTL10 in the microscopy chambers as the glycylation reaction was ongoing in the presence of ATP and glycine. These experiments showed that TTL10 recruitment to monoglycylated microtubules quickly peaks after addition of TTL10 into the chamber and then decreases, as the polyglycylation reaction proceeds, to a level close to that observed for unmodified microtubules (**Figures 4C** [↗](#), **4D** [↗](#)). In contrast, when this assay is performed in the absence of glycine, precluding the elongation of polyglycine chains, TTL10 binding to monoglycylated microtubules plateaus and does not decrease with time (**Figure 4D** [↗](#)). TTL10 binding to unmodified microtubules was close to background regardless of whether free glycine

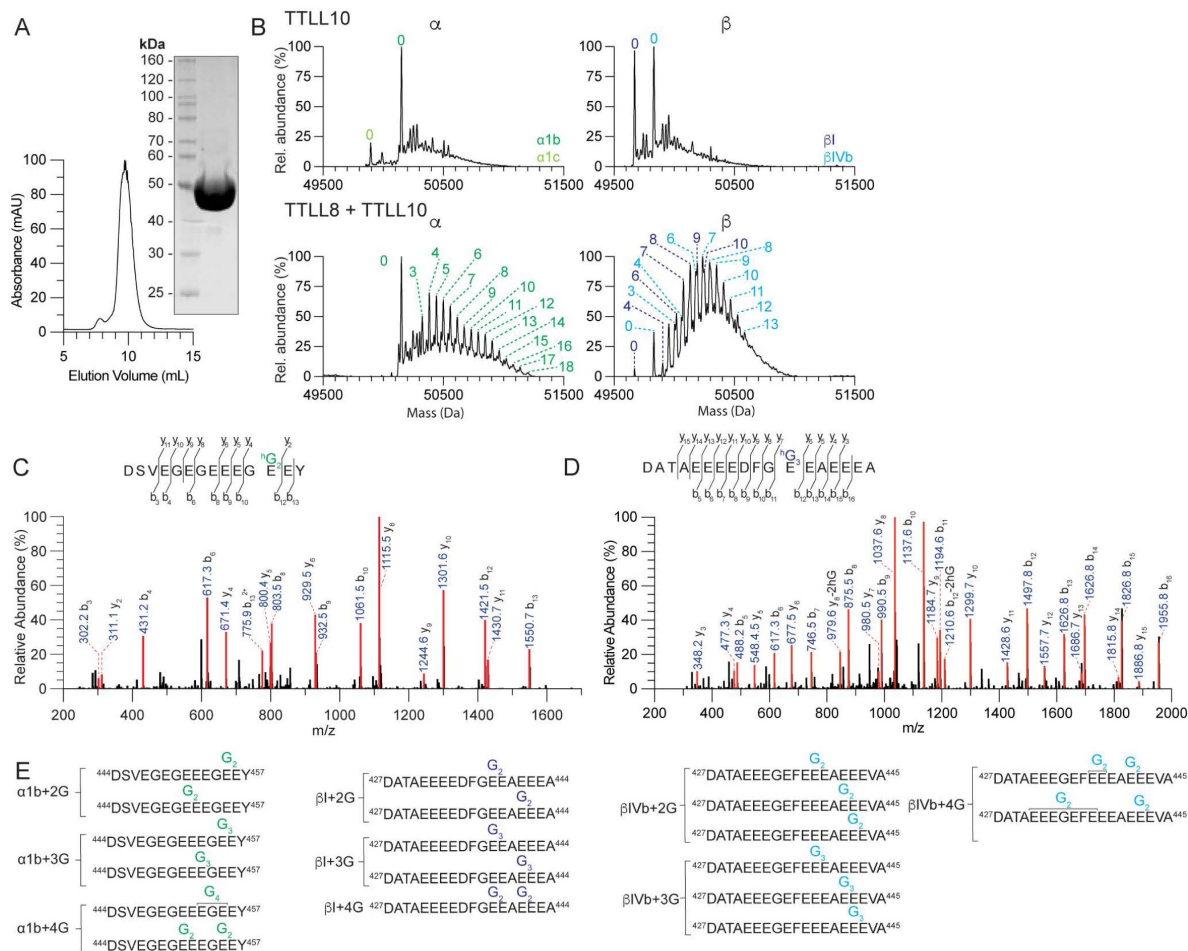


Figure 2.

TTLL10 is exclusively a tubulin polyglycylase that can only elongate existing branch-point glycines.

(A) Size exclusion column elution profile and Coomassie-stained SDS-PAGE gel of recombinant purified TTLL10. (B) Deconvoluted α - and β -tubulin mass spectra from unmodified human microtubules incubated with TTLL10 alone or with TTLL8 followed by incubation with TTLL10. The weighted mean of the number of glycines is denoted as in **Figure 1**. (C-D) Representative MS/MS spectra from two independent experiments for polyglycylated α - (C) and β -tubulin isoforms (D). C-terminal tail peptide fragments proteolytically released from microtubules glycylation by TTLL8 and TTLL10. For additional spectra see **Figure S5**. (E) List of TTLL10 polyglycylation sites identified by MS/MS sequencing.

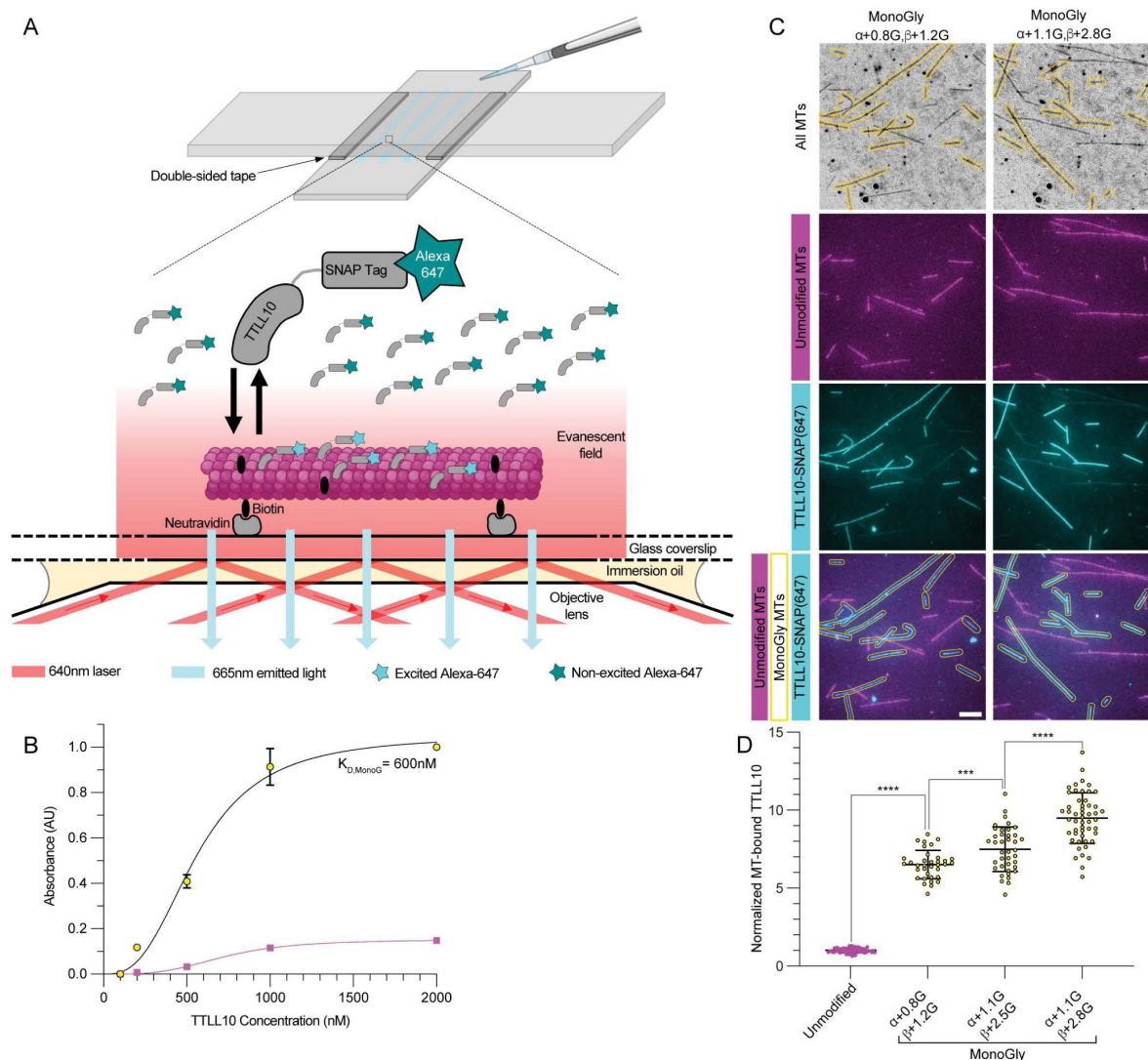


Figure 3

TLL10 requires monoglycylation for robust microtubule binding.

(A) Diagram of TIRF-based microtubule binding assays (B) Binding curve of TLL10-SNAP(Alexa647) to unmodified microtubules, shown in magenta, and monoglycylated microtubules ($\alpha + 1.7\text{G}$, $\beta + 2.0\text{G}$), shown in yellow with black outline. It was not possible to reach saturation in the reaction with unmodified microtubules. (C) TLL10-SNAP(Alexa647) (cyan) association with unmodified microtubules (Hi-Lyte 488-labeled, magenta) and monoglycylated microtubules (unlabeled, yellow contour). The weighted mean of the number of glycines denoted as in [Figure 1](#). For LC-MS of microtubules used in this assay see [Figure S6](#). Assays performed at 500 nM TLL10. Scale bar, 5 μm . (D) Quantification of TLL10-SNAP(647) recruitment to differentially monoglycylated microtubules. Values normalized to unmodified microtubules; n = 66, 35, 39, and 51 microtubules from 14, 3, 5, and 6 independent experiments for the unmodified, $\beta+1.2\text{G}$, $\beta+2.5\text{G}$, and $\beta+2.8\text{G}$ conditions, respectively. Statistical significance determined by Welch's t-test, with $p < 0.0001$ (****) or $p < 0.001$ (***).

was present or absent. These results indicate that the gradual loss of TTLL10 from the microtubule is caused by its polyglycylation activity, consistent with our results examining TTLL10 binding to differentially polyglycylation microtubules (**Figures 4A**, **4B**). Polyglycine as well as polyglutamate chain length regulate the activity of effectors such as kinesin (29) and microtubule severing enzymes (24, 30). Therefore, chain length regulation is important for controlling cellular responses. While glutamylation levels are maintained by the balance between TTLL glutamylases, which add glutamate chains (31), and CCP carboxypeptidases which remove them (32–36), an enzyme that processes glycine chains or branches has yet to be identified, raising the question of how polyglycine chain lengths are regulated in cells. The inhibition of TTLL10 binding by tubulin polyglycylation that we uncovered suggests a possible mechanism for limiting polyglycine chains lengths on tubulin, independent of a deglycylation enzyme.

Glutamylation increases TTLL10 recruitment to microtubules

Axonemal microtubules are abundantly glutamylated. Glutamylation appears during cilia development first, followed by glycylation (12, 13), indicating that in this scenario glycylation acts on pre-glutamylated microtubule substrates. TTLL6, an α -tubulin specific glutamyl elongase (28, 31, 37) localizes to cilia where it is required for their biogenesis and for normal beat patterns (12, 38). We therefore sought to determine the effect of glutamylation by TTLL6 on TTLL10 microtubule binding. For this we generated microtubules that were either only monoglycylated or both monoglycylated and polyglutamylated. Our mass spectra as well as Western blots indicate that the tubulin purified from tSA201 cells has no detectable glutamylation (**Figures 1A**, **S8**). To ensure the exact same monoglycylation levels in our assays, we first monoglycylated the microtubules. Then, we divided the sample in two: one half was glutamylated by incubation with TTLL6, while the other half was incubated with TTLL6 in the absence of glutamate (Materials and Methods). Thus, both samples had the exact same monoglycylation levels ($\langle n^G \rangle_\alpha \sim 0.1$, $\langle n^G \rangle_\beta \sim 0.8$) and differed only in their glutamylation status. TIRF-based assays with these substrates (**Figure 5A**) showed that polyglutamylation by TTLL6 ($\langle n^E \rangle_\alpha \sim 18$, $\langle n^E \rangle_\beta \sim 4$) results in a ~ 1.8 -fold increase in TTLL10 recruitment to monoglycylated microtubules (**Figure 5B**, **Figure S9**). Glutamate chains as long as ~ 17 glutamates have been documented on α -tubulin from axonemes (39). The stimulation of microtubule binding by polyglutamylation also applied to polyglycylated microtubules: TTLL10 binding to polyglycylated microtubules that were also glutamylated was ~ 3.2 -fold higher than to microtubules that were only polyglycylated (**Figure 5B**). Thus, glutamylation enhances the recruitment of TTLL10 to microtubules. Polyglycylation appears later in cilia development, after polyglutamylation. Given the moderate binding affinity of TTLL10 for monoglycylated microtubules (apparent $k_d \sim 600$ nM) and the low expression levels of TTLL10 (12), we speculate that TTLL10 is not recruited effectively to microtubules until they have been glutamylated first, ensuring the sequential addition of these modifications during cilia maturation.

In conclusion, using *in vitro* reconstitution we demonstrate that TTLL8 is strictly a glycylation initiator that catalyzes the addition of monoglycines at multiple positions on both α and β -tubulin tails, while TTLL10 is exclusively a tubulin glycylation elongase that catalyzes the addition of glycines only to pre-existing glycine branches on either α or β -tubulin. We note that TTLL glycylation enzymes seem to be more promiscuous in their substrate recognition, unlike TTLL glutamylases which show stronger preference for α or β -tubulin tails (28, 31, 40, 41). This difference in substrate stringency is consistent with the larger diversification of TTLL glutamylases (nine total identified in vertebrates so far), compared to glycylation enzymes. TTLL10 binding to microtubules requires priming of the tubulin tail with monoglycylation, establishing a hierarchy of enzyme recruitment to the microtubule. TTLL10 binding is progressively inhibited by polyglycylation, proportional with polyglycine chain length, suggesting an autonomous mechanism of polyglycine chain length control. TTLL10 recruitment to microtubules is stimulated by TTLL6 polyglutamylation, demonstrating the interplay between these two modifications. Because polyglycylation inhibits the association of TTLL10 with the microtubule, the stimulatory effect of glutamylation on TTLL10

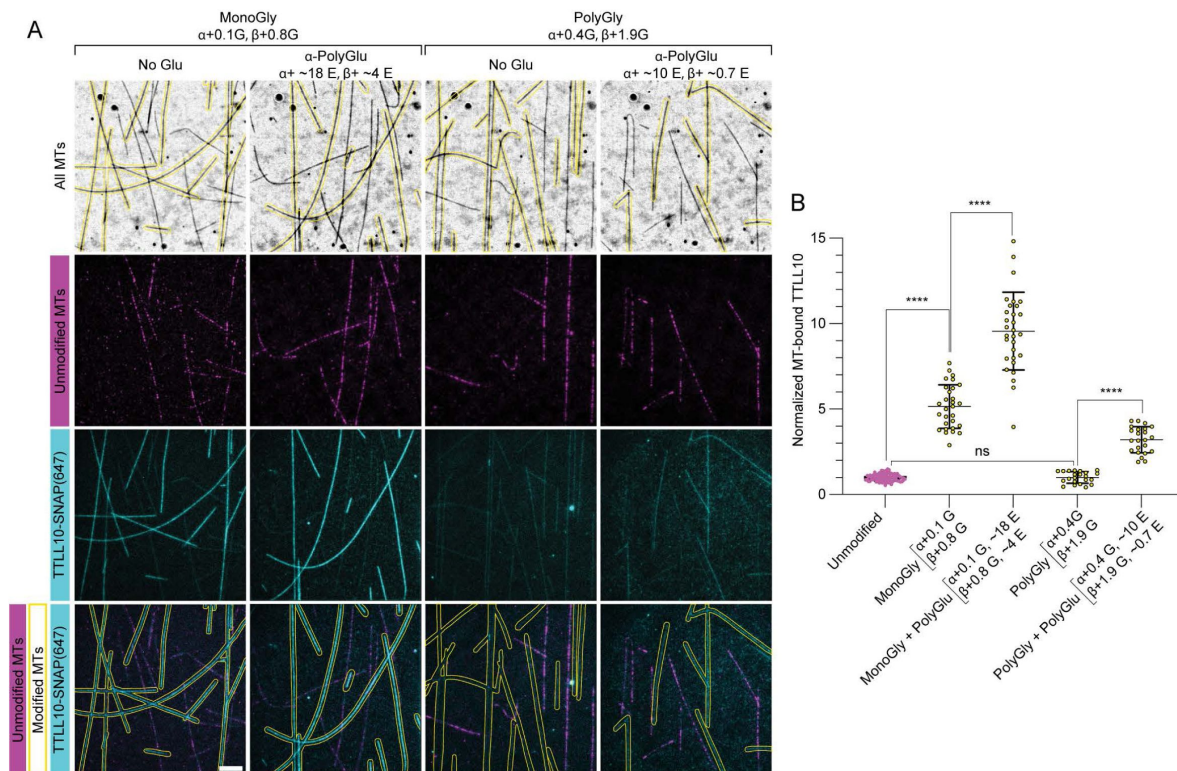


Figure 5

Polyglutamylation by TTLL6 enhances TTLL10 microtubule association.

(A) TTLL10-SNAP(Alexa647) (cyan) association with unmodified microtubules (HiLyte 488-labeled, magenta) and differentially modified microtubules (unlabeled, outlined in yellow): monoglycylated, monoglycylated/polyglutamylated, polyglycylated only, and polyglycylated/polyglutamylated. Monoglycylation added by TTLL8, polyglycylation by TTLL10 and polyglutamylation by TTLL6 (Materials and Methods). The weighted mean of the number of glycines denoted as in **Figure 1**. Glutamylation levels for glycylated/glutamylated microtubules are approximate and estimated by western blot (**Figure S9**) because quantification from LC-MS spectra of these dual-modified microtubules was not possible due to the large number of species present. Scale bar, 5 μ m. (B) Quantification of TTLL10-SNAP(Alexa647) recruitment to modified microtubules normalized to those on unmodified microtubules; $n = 84, 29, 30, 22$, and 23 microtubules from 10, 4, 4, 3, and 3 independent experiments for monoglycylated, monoglycylated/polyglutamylated, polyglycylated, and polyglycylated/polyglutamylated microtubules, respectively. Statistical significance determined by Welch's t-test, with $p < 0.0001$ (****).

recruitment can potentially generate longer polyglycine chains on microtubules that are already glutamylated and shorter polyglycine chains on microtubules that are not glutamylated. The stimulatory effect of glutamylation does not necessarily result in increased glycylation of the same tubulin dimer that is glutamylated because TTLLs can interact with the tubulin tails of neighboring tubulin dimers on the microtubule (40 [↗](#)), and also because increased recruitment to one glutamylated tubulin subunit ensures a higher probability of rebinding at a nearby tubulin on the microtubule. These mechanisms can result in an enrichment of glycylation on the same microtubule, but not necessarily on the same tubulin dimer that carries the stimulatory polyglutamate chains on its tail. Recent work revealed that glutamylation also stimulates vasohibin (42 [↗](#)), offering a possible explanation for the high degree of overlap between detyrosination and glutamylation on subpopulations of microtubules in cells. More recently, the α -tubulin glutamylase TTLL6 was shown to read out the modification status of the β -tubulin tail on the laterally adjacent neighboring tubulin dimer, facilitating crosstalk between tubulin tails (43 [↗](#)). Our present study adds to the growing evidence of crosstalk between tubulin modifications and establishes the biochemical mechanisms that govern the syntax of the tubulin code. Lastly, a major limitation towards our mechanistic understanding of glycylation and the tubulin code overall has been the inability to generate microtubules with diverse and controlled posttranslational modifications patterns for *in vitro* reconstitution. By identifying the sites of modifications introduced by TTLLs glycylation and the interplay between their activities, we not only provide insight into the biochemical program that generates the complex *in vivo* microtubule glycylation patterns, but also a tool for generating differentially glycylation microtubules to study *in vitro* the effects of glycylation on microtubule effectors and thus shed light on the mechanism of action of the tubulin code.

Materials and Methods

Protein expression and purification

Homo sapiens TTLL8 (res. 39-585) was cloned into pCoofy28 (*Addgene plasmid* #44004 [↗](#)) by sequence-specific ligation with RecA (44 [↗](#)), and bacmid was produced by transformation of DH10EMBacY (Geneva Biotech). Baculovirus was produced by transfection of and subsequent amplification in ExpiSf9 cells (Thermo Fisher Scientific). TTLL8 was expressed in baculovirus-infected Sf9 cells (Thermo Fisher Scientific) with an N-terminal Glutathione S-transferase tag. Cells were infected at a density of 2.5×10^6 cells/mL with an MOI of 2 and then grown for 48 hrs before harvest. Cells were lysed using a microfluidizer and cell lysate was clarified by centrifugation at 45,000 rpm (235,000 rcf) for 60 minutes using a Type 45 Ti rotor (Beckman-Coulter). The fusion protein was captured using GST affinity chromatography. TTLL8 was liberated from the resin using 3C protease at a 1:500 molar ratio, further purified on a heparin column, and finally polished by size-exclusion chromatography. TTLL8 was flash-frozen in small aliquots in 50mM HEPES pH 7.0, 10mM $MgCl_2$, 200mM KCl, 2mM TCEP, 10% glycerol.

Xenopus tropicalis TTLL10 (res. 105-570) was cloned into pCoofy28 (*Addgene plasmid* #44004 [↗](#)) and bacmid was produced by transformation of DH10EMBacY (Geneva Biotech). TTLL10 was expressed and purified following the same protocol as the one described above for TTLL8. TTLL10 was flash-frozen in 20mM Tris pH 8.0, 10mM $MgCl_2$, 100mM NaCl, 2mM TCEP, 10% glycerol. TTLL10 with a C-terminal SNAP tag was expressed and purified as described above; however, the protein was conjugated to a SNAP-Surface Alexa Fluor 647 dye (New England Biosciences) after elution from the heparin column by incubation with the fluorophore on ice overnight. Excess dye was removed by size-exclusion chromatography whereupon TTLL10-SNAP(647) was flash-frozen in 20mM Tris pH 8.0, 10mM $MgCl_2$, 100mM NaCl, 2mM TCEP, 10% Glycerol.

Unmodified human tubulin from tsA201 cells was purified as previously described using a TOG1 affinity column (27 [↗](#)).

Generation of glycylation microtubules for mass spectrometric analyses

The preparation of unmodified microtubules for this study closely follows previously published protocols (27, 30). Microtubules were polymerized from 10 μ M unmodified human tubulin in BRB80 (80mM PIPES, pH 6.8, 1mM MgCl₂, 1mM EGTA) with 10% DMSO and 1mM GTP. Taxol was added to 20 μ M after 1hr, and then left to incubate overnight. Free tubulin was removed by spinning the polymerization mixture through a 60% glycerol cushion in BRB80 and the microtubules were resuspended in BRB80 with 20 μ M Taxol to a final concentration of 20 μ M. Glycylation reactions were performed at room temperature and commenced by addition of enzyme after the microtubules were evenly dispersed in reaction buffer. C¹³ glycine was used to improve separation of the modified β I and β IVb tubulin peaks by LC-MS and facilitate the identification of glycine-modified peptide fragments by MS/MS. After every glycylation reaction, bound glycylation enzyme was removed with a high salt wash by adding 300mM KCl, incubating at 37°C for 15 minutes, and subsequently spinning through a 60% glycerol cushion in BRB80 (80mM KOH-PIPES, pH 6.8, 1mM MgCl₂, 1mM EGTA). For monoglycylation reactions, unmodified human microtubules stabilized with 20 μ M taxol were incubated with TTL8 (0.2 μ M) at a 1:50 enzyme:tubulin molar ratio for 20 minutes at room temperature. For polyglycylation reactions with TTL10, taxol-stabilized, unmodified human microtubules were first monoglycylated for 4 hours by TTL8 (0.2 μ M) at a 1:20 enzyme:tubulin molar ratio, to ensure complete monoglycylation of the substrate. Bound TTL8 was removed by a salt wash as described above, whereupon the monoglycylated microtubules were subsequently treated with TTL10 (0.2 μ M) at a 1:50 enzyme:tubulin molar ratio for 20 minutes at room temperature. Bound TTL10 was removed by a second salt wash performed as described above.

Mass spectrometric analyses of glycylation reactions

The number of glycines added to α - and β -tubulin was determined by LC-MS. Reaction samples for LC-MS were prepared as described above and then quenched at 20 minutes *via* addition of 20mM EDTA on ice. Samples for LC-MS were diluted to 0.2 μ g/ μ L in cold BRB80 and then diluted 1:1 with 0.1% trifluoroacetic acid (TFA) in 20% acetonitrile. 1 μ g of microtubules were separated using a 0-70% acetonitrile gradient in 0.05% TFA at a flow rate of 0.2mL/min. The column was coupled to a 6224 ESI-TOF LC-MS (Agilent) and the spectra were deconvoluted using the Agilent MassHunter software and display a distribution of masses with consecutive glycylation species separated by 58 Da corresponding to one [¹³C]-glycine. The extent of tubulin glycylation on α - or β -tubulin was determined by calculating the weighted average of peak intensities for each tubulin species present.

Samples for MS/MS analyses were reduced with 5 mM Tris(2-carboxyethyl)phosphine (TCEP) hydrochloride at room temperature for 1hr, alkylated with 10 mM N-Ethylmaleimide for 10 min. Each sample was divided into 2 aliquots and digested with trypsin (Trypsin Gold, Mass Spectrometry Grade, Promega) and AspN (Sequencing Grade, Roche) separately. Trypsin and AspN digestion was performed with 1:15 enzyme:sample (w/w) at 37 °C for 12 hr. Digested samples were desalted using μ Elution HLB plate (Waters). Data acquisition was performed on a system where an Ultimate 3000 HPLC (Thermo Scientific) was coupled to an Orbitrap Lumos mass spectrometer (Thermo Scientific) via an Easy-Spray ion source (Thermo Scientific). For each LC-MS/MS run, 0.5 μ g digests were injected. Peptides were separated on an ES902 Easy-Spray column (Thermo Scientific). The composition of mobile phases A and B was 0.1% formic acid in HPLC water, and 0.1% formic acid in HPLC acetonitrile, respectively. The mobile B amount was increased from 3% to 27% in 66 minutes at a flowrate of 300 nL/min. Thermo Scientific Orbitrap Lumos mass spectrometer was operated in data-dependent mode. The MS1 scans were performed in orbitrap with a resolution of 120K at 200 m/z and a mass range of 400-1500 m/z. Collision-induced dissociation (CID) method was used for MS2 fragmentation. MS2 scans were conducted in ion trap. The precursor ion isolation width was 1.6 m/z, and the dynamic exclusion window was 4 sec.

Database search was performed using Mascot against Sprot Human database. The mass tolerances for precursor and fragment were set to 10 ppm and 0.6 Da, respectively. Up to three missed cleavages were allowed for data obtained from trypsin digestion, and up to four missed cleavages were allowed for AspN data. NEM on cysteines was set as fixed modification. Variable modifications include Oxidation (MW), Met-loss (Protein N-term), Acetyl (Protein N-term), 1hG(G), 2hG(G), 3hG(G), and 4hG(G). Peptides matched with heavy glycine modification were manually curated.

Generation of differentially glycylation microtubules for microscopy assays

Unmodified human microtubules were polymerized from 10 μ M human unmodified tubulin with 1.5% biotinylated brain tubulin in BRB80 by addition of 10% DMSO and incubation for 1hr before adding 10 μ M taxol. These microtubules were modified with TTL8 alone or with both TTL8 and TTL10. The mean number of glycines was determined by LC/MS. Specifically, to prepare differentially monoglycylated microtubules, 10 μ M unmodified microtubules were incubated at room temperature with 1:20 TTL8 for 1hr ($\langle n^G \rangle_\alpha \sim 0.8$, $\langle n^G \rangle_\beta \sim 1.2$), 2hr ($\langle n^G \rangle_\alpha \sim 1.1$, $\langle n^G \rangle_\beta \sim 2.5$), and 4hr ($\langle n^G \rangle_\alpha \sim 1.1$, $\langle n^G \rangle_\beta \sim 2.8$). To prepare differentially polyglycylated microtubules, unmodified microtubules were first modified at room temperature for 4hrs with 1:20 TTL8 ($\langle n^G \rangle_\alpha \sim 1.1$, $\langle n^G \rangle_\beta \sim 2.8$). After removing TTL8 using a salt wash as described above and subsequently spinning through a glycerol pad, these microtubules were treated with 1:20 TTL10 for 2 ($\langle n^G \rangle_\alpha \sim 3.0$, $\langle n^G \rangle_\beta \sim 4.3$), 5 ($\langle n^G \rangle_\alpha \sim 3.3$, $\langle n^G \rangle_\beta \sim 5.4$), and 15 minutes ($\langle n^G \rangle_\alpha \sim 3.3$, $\langle n^G \rangle_\beta \sim 7.7$). To prepare microtubules that were glycylation and glutamylated, unmodified microtubules were first modified with TTL8 at room temperature for 1hr at a molar ratio of 1:50 ($\langle n^G \rangle_\alpha \sim 0.1$, $\langle n^G \rangle_\beta \sim 0.8$) whereupon TTL8 was removed by using a salt wash and spinning through a glycerol pad as described above. A portion of these microtubules were then modified at room temperature for 1hr with 1:50 TTL10 ($\langle n^G \rangle_\alpha \sim 0.4$, $\langle n^G \rangle_\beta \sim 1.9$) whereupon TTL10 was removed by using a salt wash and spinning through a glycerol pad as described above. The monoglycylated microtubules and the polyglycylated microtubules were then split in half, with one half left untreated and the other half treated at room temperature overnight with 1:10 TTL6 to obtain long polyglutamate chains primarily on α -tubulin (28⁴⁵), after which TTL6 was removed through another salt wash and by spinning through a glycerol pad as described above. We note that we first glycylation and then glutamylated to ensure that we have the exact same glycylation levels for the unglutamylated and glutamylated microtubules. Glutamylated levels were estimated by western blot for glycylation/glutamylated microtubules, as it is not possible to quantify modification levels using LC-MS due to the large number of species present in the product mixture. Blots were stained first with a mixture of 1:10,000 rabbit anti-polyE antibody (clone IN105, Adipogen) and 1:10,000 mouse anti- α -tubulin antibody (clone DM1A, Sigma-Aldrich) for 1hr, washed, stained with a mixture of 1:10,000 IRDye® 680RD goat anti-mouse antibody (LI-COR) and 1:10,000 IRDye® 800CW goat anti-rabbit antibody (LI-COR), washed again, and then imaged on an Odyssey imager (LI-COR).

For **Figures S3** and **S8**, microtubule samples were subjected to SDS-PAGE and Western Blot transfer. 200 ng of tubulin was loaded for each unmodified and glutamylated and glycylation sample. The membranes were blotted for modifications with the following antibodies: glutamylated with mouse GT335 antibody (AdipoGen Cat# AG-20B-0020-C100) at dilution 1:4000, mono-glycylation and bi-glycylation with rabbit Gly-pep1 (AdipoGen Cat# AG-25B-0034-C100) at dilution 1:2000, polyglycylation with the poly-Gly rabbit polyclonal antibody at dilution 1:4000 (45⁴⁵). Tubulin loading controls were detected with mouse anti- α -tubulin antibody (DM1A) (Acbam Cat# ab7219) at dilution 1:4000. Western Blots were visualized with secondary antibodies: IRDye® 800CW Goat anti-Rabbit IgG Secondary Antibody (LiCor Cat# P/N: 926-32211) and IRDye® 680RD Goat anti-Mouse IgG Secondary Antibody (LiCor Cat# P/N: 926-68070) at dilution 1:18000. All antibody dilutions were made in 4% milk in phosphate buffered saline with 0.2% tween.

TIRF based microtubule binding assays

Flow chambers were constructed of plasma-cleaned and silanized glass slides and coverslips as previously described (46). Chambers were coated with neutravidin and then washed with BRB80 with 2mg/mL casein and 20μM taxol followed by BRB80 with 1% Pluronic F-127 and 20μM taxol. Microtubules were immobilized in the chambers and washed again with BRB80 with 2mg/mL casein and 10μM taxol followed by BRB80 with 1% Pluronic F-127 and 10μM taxol and finally equilibrated in binding buffer (60mM KCl, 10mM NaCl, 40mM PIPES pH 6.8, 2mM Tris pH 8.0, 20mM Glucose, 1.5mM MgCl₂, 1.5mM TCEP, 1mM ATP 0.5mM EGTA, 1% Pluronic, 10μM Taxol). Taxol-stabilized human unmodified microtubules assembled with 1.5% biotinylated tubulin (Cytoskeleton, Inc. #T333P) and 1% HiLyte FluorTM 488-labeled tubulin (Cytoskeleton, Inc. #TL488M) were used as an internal control in all chambers for all assays. Microtubule binding assays were performed at 500nM TTLL10-SNAP(647) and were supplemented with oxygen scavenger mix to remove free oxygen from solution (7.5μM glucose oxidase, 0.5μM catalase). TTLL10-SNAP(647) was perfused into the chamber during image acquisition. Background-corrected line scan intensities were measured over time using Fiji and normalized to microtubule length and unmodified microtubules which were used as an internal standard in each chamber. Peak binding intensities were used for all microtubules.

Fluorescence images were acquired using an inverted total internal reflection fluorescence (TIRF) microscope (Nikon Ti-E with TIRF attachment) equipped with an iXON3-897 EMCCD camera (Andor). For TIRF assays, the excitation light was provided by a 488nm (Coherent Sapphire) or 647nm (Coherent Cube) laser set to 20mW before being coupled into the microscope *via* an optical fiber to the Ti-TIRF arm on the microscope. Light was delivered to the sample through the TIRF arm and directed towards a 100x 1.49 NA TIRF objective (Nikon CFI Apo TIRF 100x). Emission light was split using a dichroic beamsplitter FF640-FDi02 (Semrock) and further filtered using a FF01-550/88 filter (Semrock) on the 488 channel or a BLP01-647R filter on the 647nm channel. Prior to perfusing TTLL10, an image of all microtubules was taken using interference reflection microscopy (IRM) using an ORCA-Flash4.0 CMOS camera (Hamamatsu) and an image of the unmodified HiLyte 488-labeled, internal reference microtubules was taken by TIRF at 100ms exposure with excitation at 488nm. TTLL10-SNAP(647) binding during and after perfusion was imaged by TIRF with 100ms exposure at 2s intervals with excitation at 647nm.

Data availability

All plasmids used in this study are available from the corresponding author upon request and source data will be provided for each figure.

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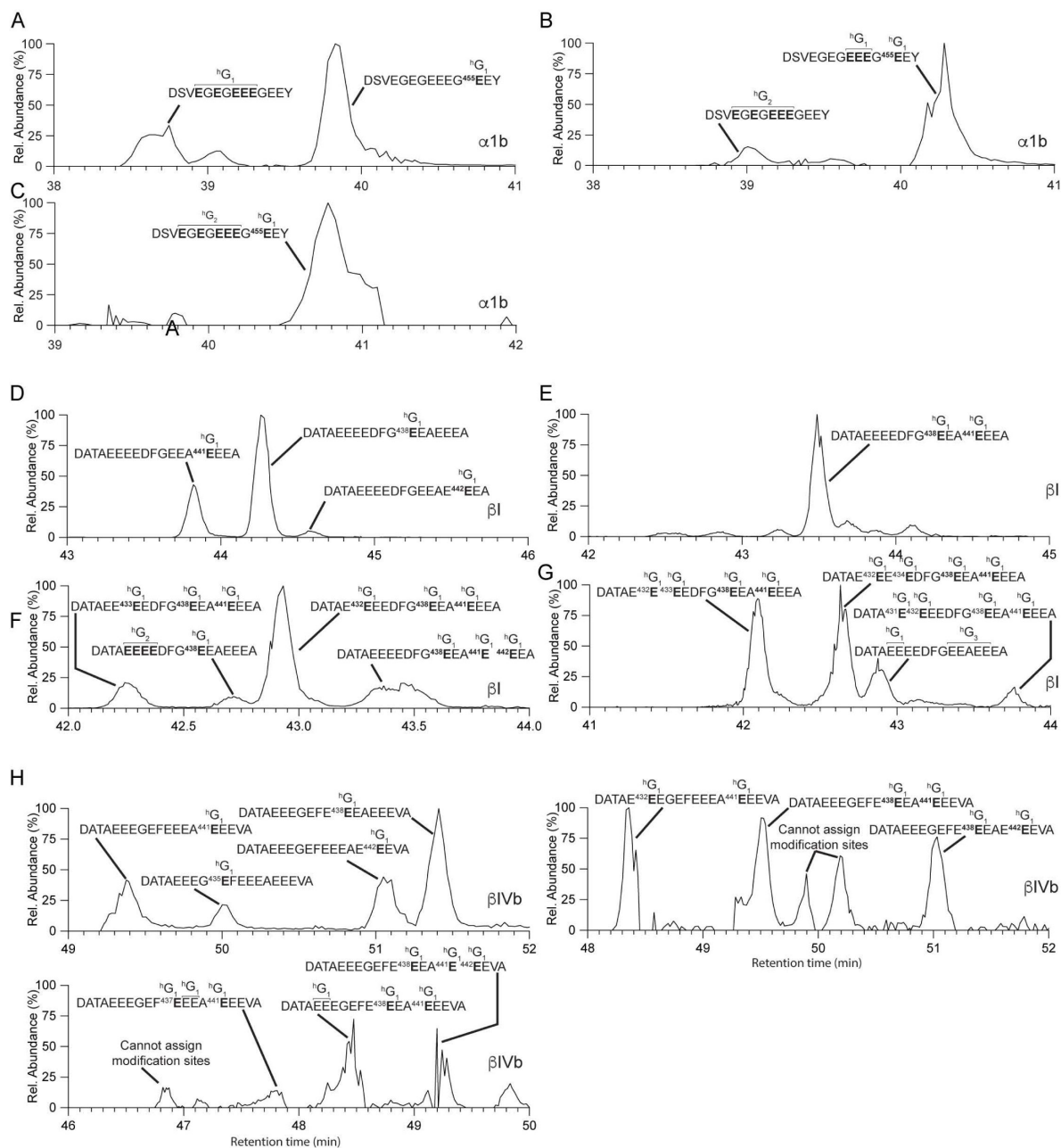


Figure S1.

Extracted-ion chromatograms for monoglycylated α1b, βI, and βIVb tubulin tails modified by TTLL8.

(A-C). Extracted-ion chromatograms for monoglycylated α1b-tail peptides with one monoglycine branch (A), two monoglycine branches (B) and three monoglycine branches (C). (B-G). Extracted-ion chromatograms for monoglycylated βI-tail peptides with one monoglycine branch (D), two monoglycine branches (E), three monoglycine branches (F) and four monoglycine branches (G). (H). Extracted-ion chromatograms for monoglycylated βIVb-tail peptides. Tubulin tail peptides were proteolytically released from microtubules incubated with TTLL8.

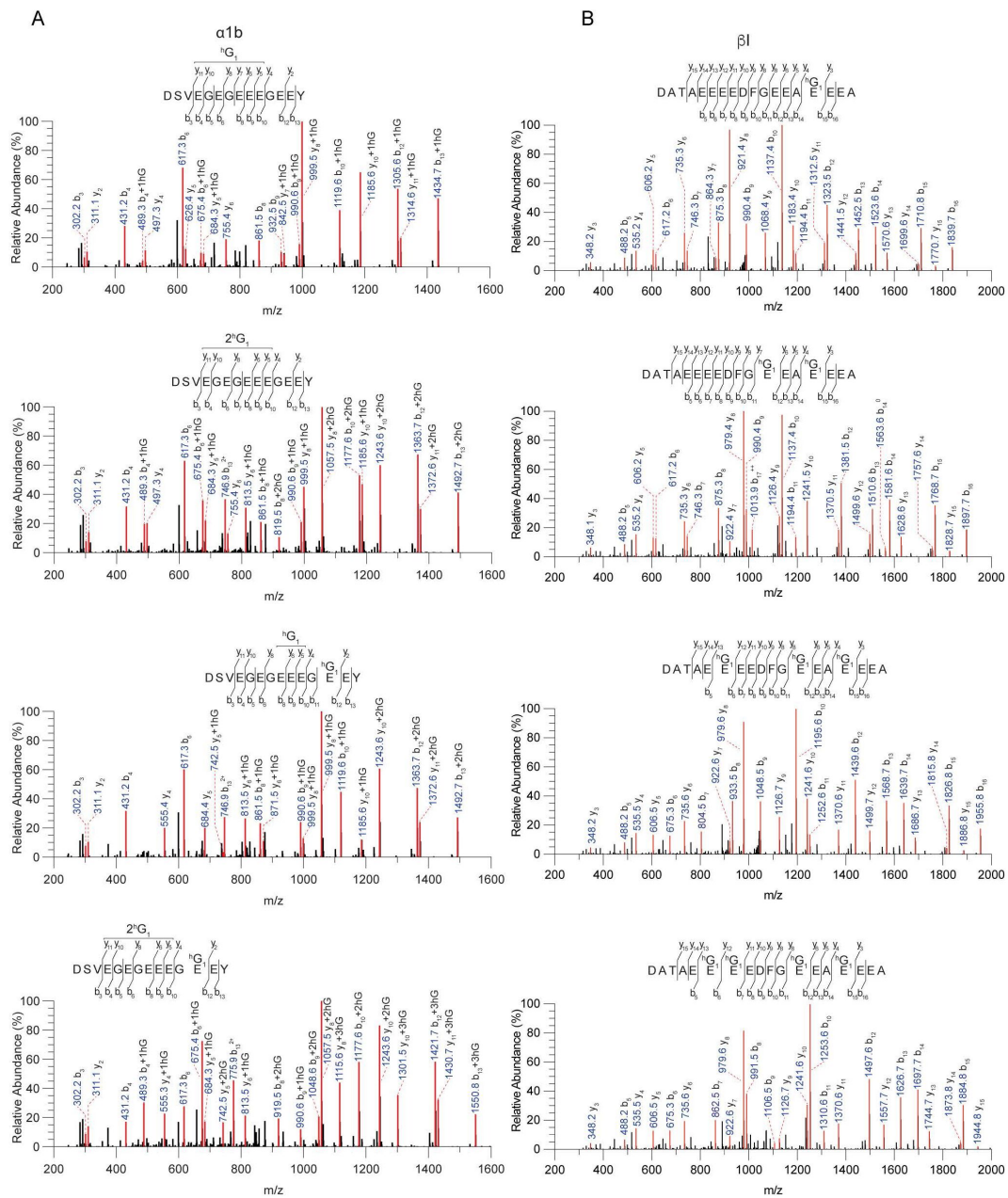


Figure S2.

MS/MS spectra for monoglycylated tubulin C-terminal tail peptides proteolytically excised from microtubules incubated with TTL8.

(A, B). MS/MS spectra for monoglycylated α 1b-peptides (A) and β I-tubulin C-terminal tail peptides (B).

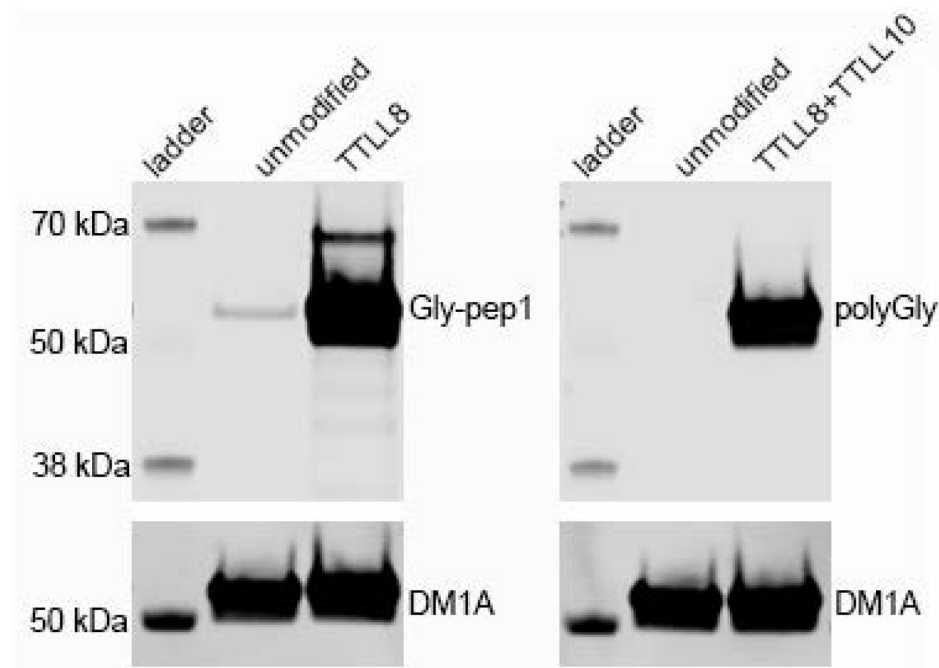


Figure S3.

Western blot analysis of glycylation in tubulin purified from tSA201.

0.2 μ g of purified tubulin were loaded for each unmodified and glycylation sample. Glycylation was detected with the Gly-pep1 and anti-polyG antibodies (Materials and Methods). We note a weak reactivity to the gly-pep1 antibody for the tubulin purified from tSA201 cells. It is unclear whether the antibody detects a very small proportion of mono- or bi-glycylation tubulin in this sample or the signal is due to a low affinity interaction of the antibody for the unmodified tubulin tail which is visible at these tubulin loading levels. Neither our LC-MS or MS/MS data of the tSA201 tubulin detected any glycylation on the intact tubulin, or tubulin tails, respectively. The higher molecular weight band is from TTLL8 which self-modifies during expression. The tSA201 tubulin shows no reactivity against the polyglycylation antibody.

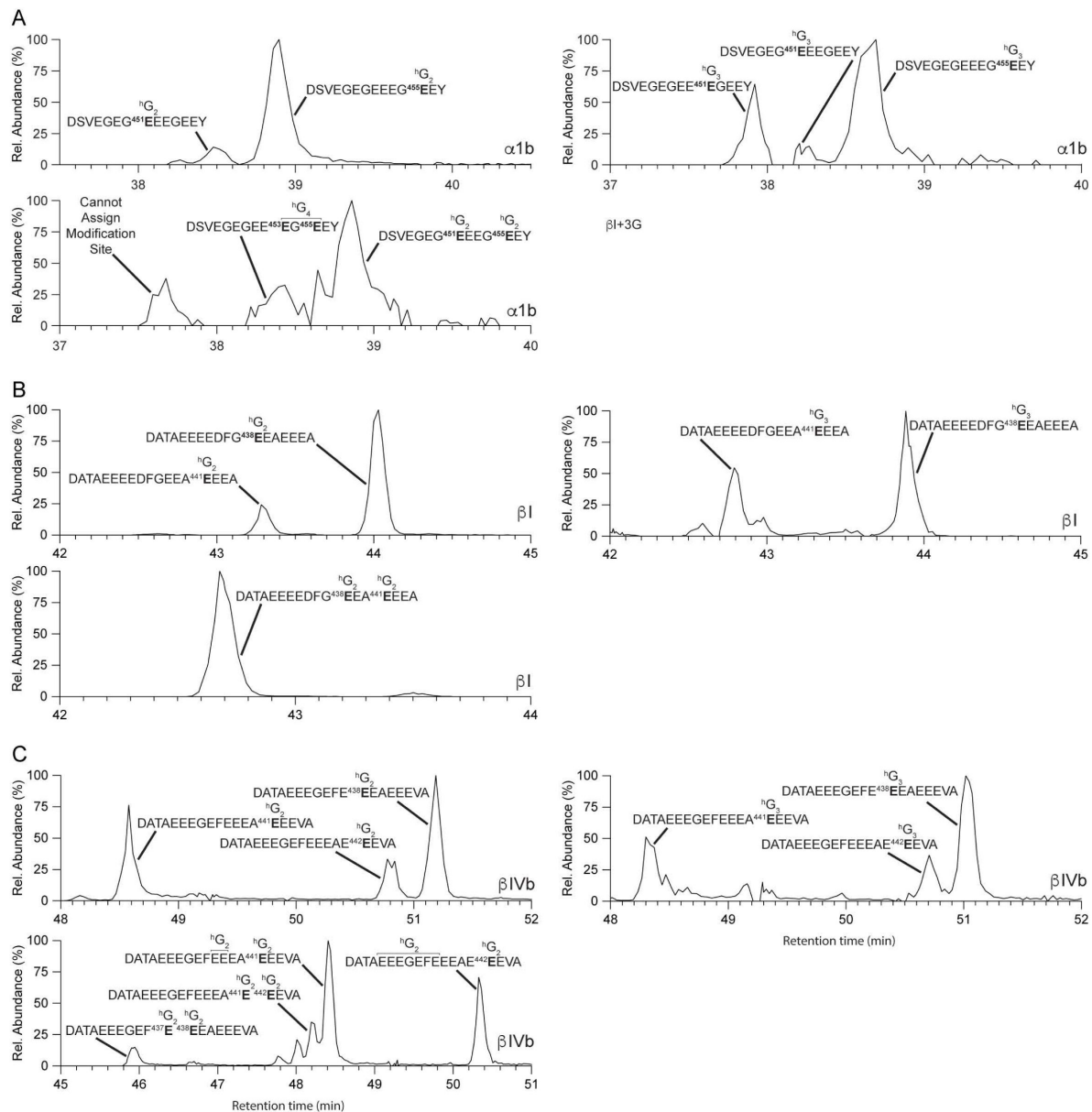


Figure S4.

Extracted-ion chromatograms for polyglycylated tubulin C-terminal tail peptides proteolytically excised from monoglycylated microtubules incubated with TTL10.

(A-C). Extracted-ion chromatograms for polyglycylated α1b-peptides (A), βI-peptides (B) and βIVb-tubulin (C) C-terminal tail peptides. The subscript in G_i indicates the length of the polyglycine chain.

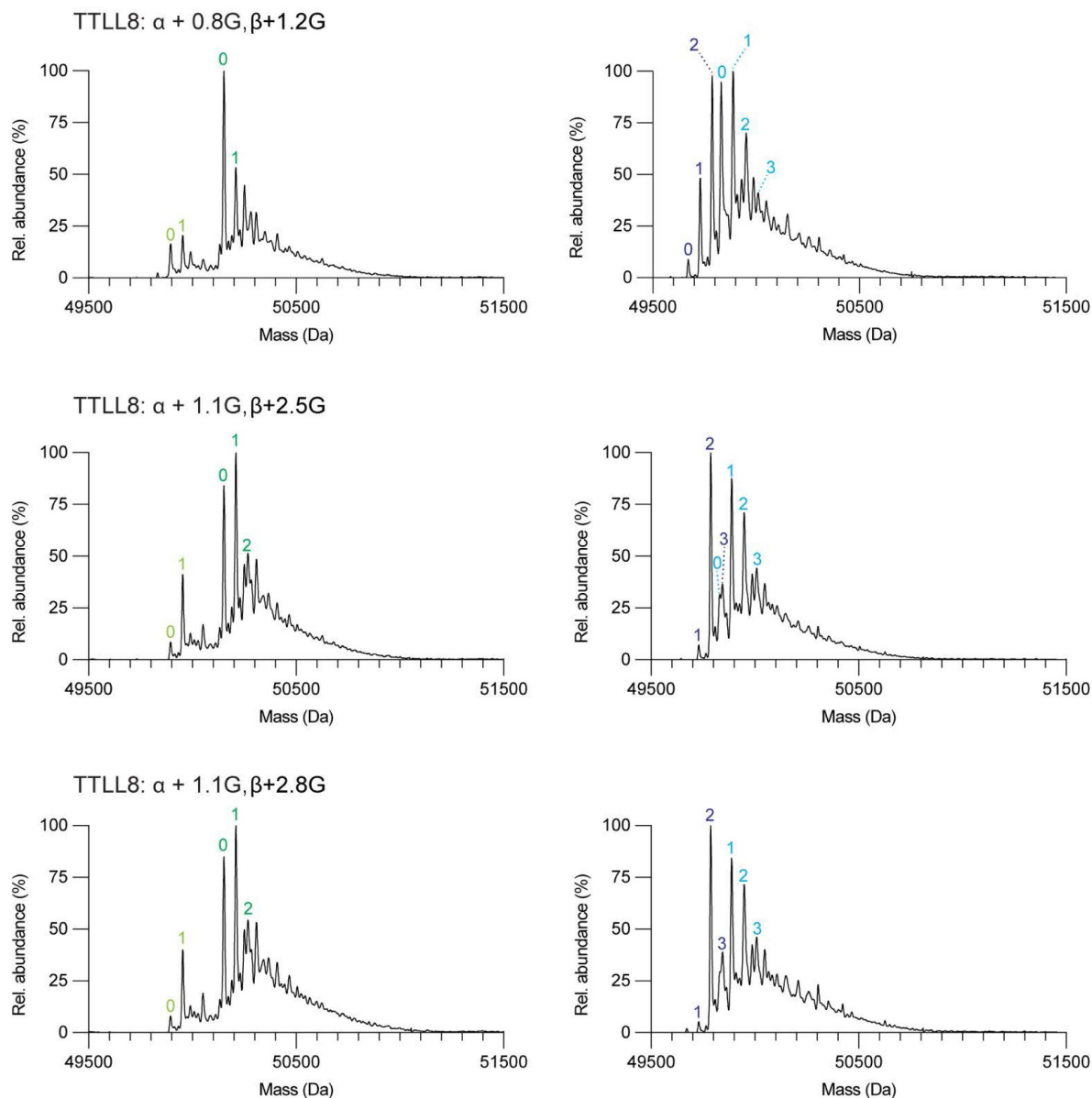


Figure S6.

LC-MS of differentially monoglycylated microtubules used in assays shown in Figure 3 [↗](#).

The weighted mean of the number of glycines ($\langle n^G \rangle$) added to α - and β -tubulin are denoted $\alpha + \langle n^G \rangle$; $\beta + \langle n^G \rangle$. The number of posttranslationally added glycines is indicated in green (for α -tubulin isoforms) and blue (β -tubulin isoforms) on the spectra.

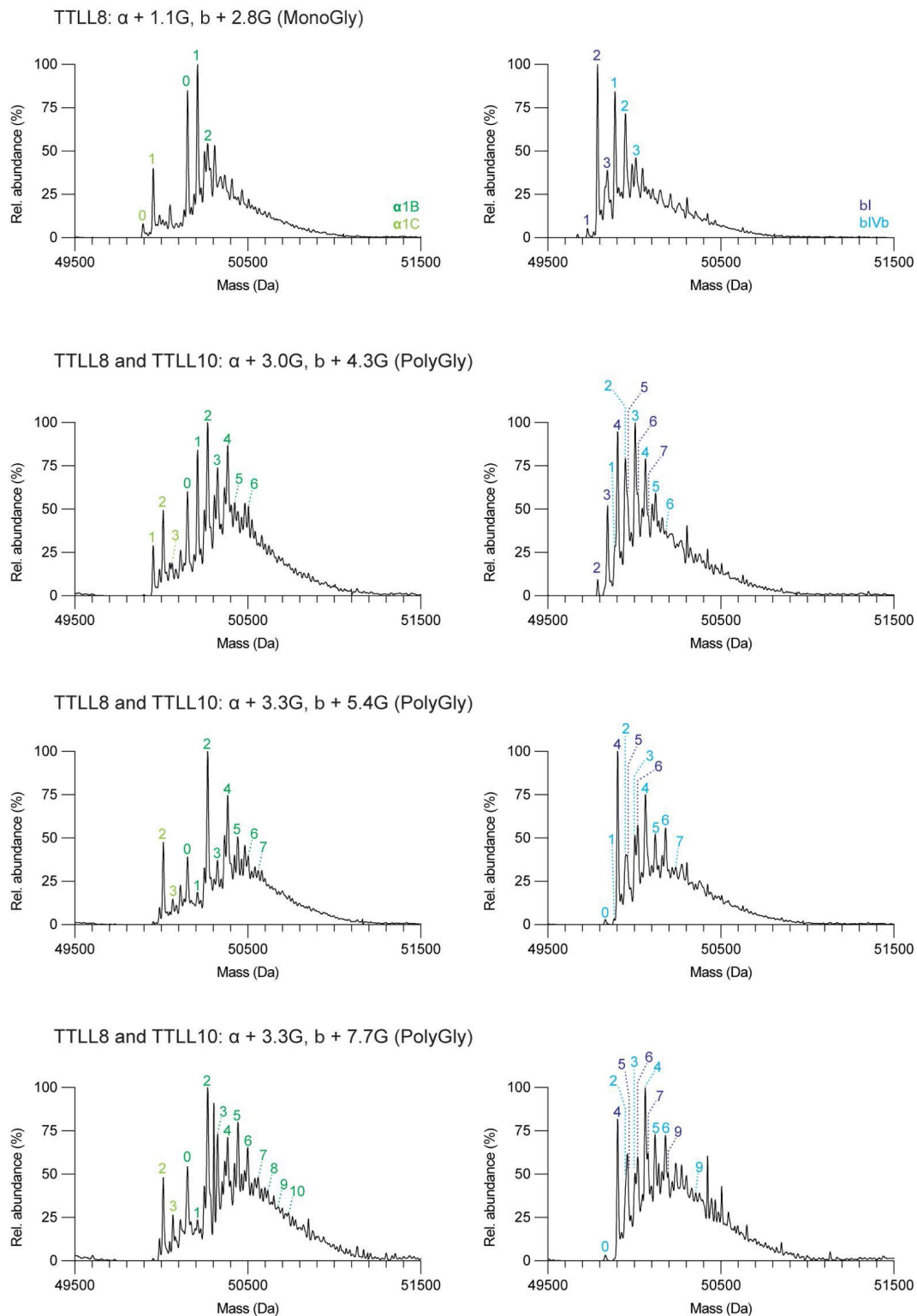


Figure S7.

LC-MS of monoglycylated and polyglycylated microtubules used in binding assays shown in Figure 4.

The weighted mean of the number of glycines ($\langle n^G \rangle$) added to α - and β -tubulin are denoted $\alpha + \langle n^G \rangle$; $\beta + \langle n^G \rangle$. The number of posttranslationally added glycines is indicated in green (for α -tubulin isoforms) and blue (β -tubulin isoforms) on the spectra.

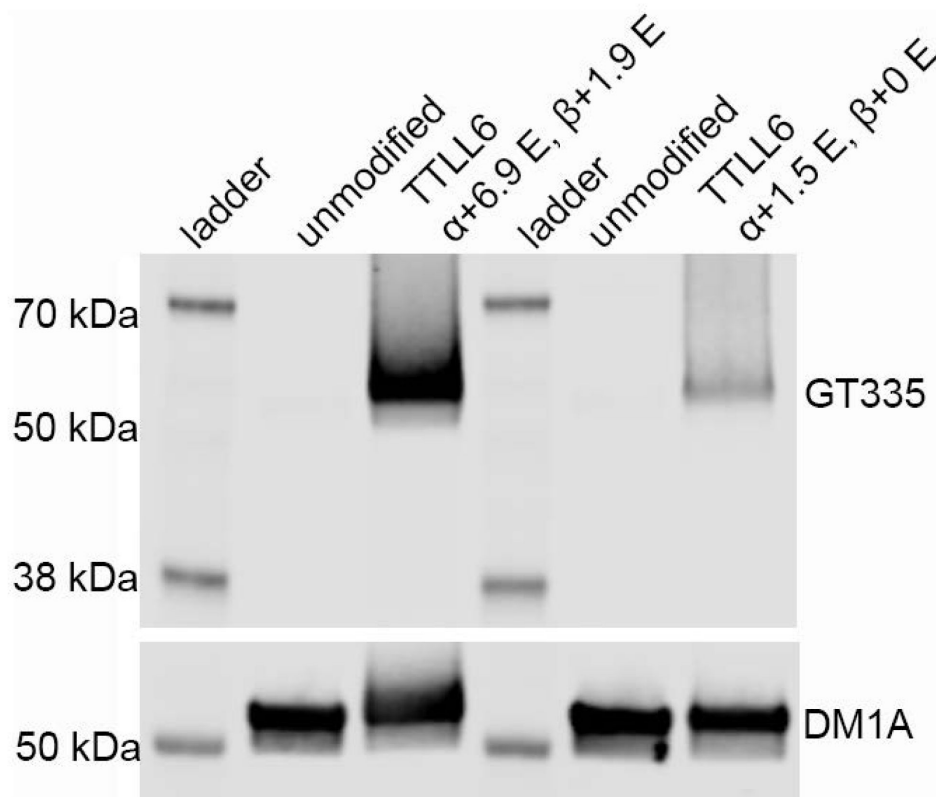
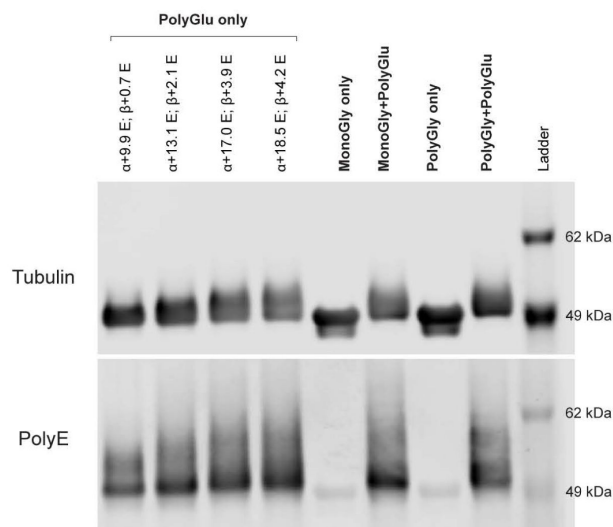


Figure S8.

Western blot analysis of glutamylation in tubulin purified from tSA201 cells.

0.2 µg of tubulin was loaded for each unmodified and glutamylated sample. Glutamylation was detected with the GT335 antibody (Materials and Methods). No signal is detectable in the unmodified tSA201 purified tubulin. The MS/MS analysis of this tubulin did not identify any glutamylated peptides, either. In contrast, the tSA201 tubulin enzymatically glutamylated *in vitro* shows strong signal that increases with glutamylation level. Number of posttranslationally added glutamates indicated and determined from LC-MS measurements. The glutamylated tubulin runs slightly higher due to the changes in electrophoretic mobility.

A



B

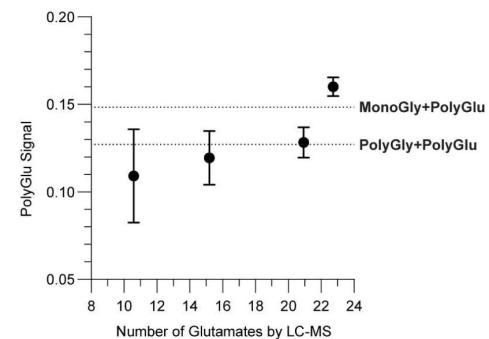


Figure S9.

Estimation of glutamylation levels of monoglycylated and polyglycylated microtubules used in microtubule binding assays shown in Figure 5.

(A) Western blot of polyglutamylated and monoglycylated or polyglycylated microtubules used in microtubule binding assays shown in Figure 5. From the left, first four lanes, TTLL6 polyglutamated microtubules with listed mean glutamate numbers determined from LC-MS spectra of intact microtubules. These were used to calibrate the polyglutamate signal detected with an anti-poly-E antibody (clone IN105, Adipogen; Materials and Methods); Subsequent four lanes, dually modified microtubules (glycylated with TTLL8 or TTLL8+10, and glutamylated with TTLL6) used in the assays shown in Figure 5; (B) Poly-E signal as a function of total glutamate numbers on α - and β -tubulin. The signal for the two dually modified species (monoglycylated + polyglutamylated and polyglycylated + polyglutamylated) used in the TIRF-based assays is shown with a discontinuous line.

Additional information

Author contributions

SWC expressed and purified proteins, did LC-MS and microscopy assays; YL did MS/MS, JOS did microscopy assays together with SWC. SWC and JOS did image analysis. CK did initial expression and purification of TTLL8. SWC, YL, JOS analyzed data. SWC and ARM planned experiments, discussed data and wrote the manuscript. ARM conceived the project and supervised research. All authors reviewed and approved the manuscript.

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Reviewer #1 (Public review):

Summary:

In their current study, Cummings et al have approached this fundamental biochemical problem using a combination of purified enzyme-substrate reactions, MS/MS and microscopy in vitro to provide key insights into the hierarchy of generating polyglycylation in cilia and flagella. They first establish that TTLL8 is a monoglycylase, with the potential to add multiple mono glycine residues on both α - and β -tubulin. They then go on to establish that the monoglycylation is essential for TTLL10 binding and catalytic activity, which progressively reduces as the level of polyglycylation increases. This provides an interesting mechanism of how level of polyglycylation is regulated in the absence of a deglycylase. Finally, the authors also establish that for efficient TTLL10 activity, it is not just monoglycylation, but also polyglutamylolation that is necessary, giving a key insight into how both these modifications

interact with each other to ensure there is a balanced level of PTMs on the axonemes for efficient cilia function.

Strengths:

The manuscript is well written, and experiments are succinctly planned and outlined. The experiments used provide the conclusions to what the authors were hypothesising and provide some new novel possible mechanistic insights into the whole process of regulation of tubulin glycylation in motile cilia.

Weaknesses:

There were some weaknesses in the initial submission of the manuscript, but the authors have addressed these in their revised version either by giving clear explanations in the text or through additional experiments.

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Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

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Strengths:

The manuscript is well-written, and experiments are succinctly planned and outlined. The experiments were used to provide the conclusions to what the authors were hypothesising and provide some new novel possible mechanistic insights into the whole process of regulation of tubulin glycylation in motile cilia."

We thank the reviewer for their support of our study and recognition of its importance to understanding microtubule glycylation and its regulation.

"The initial part of the manuscript where the authors discuss about the requirement of monoglycylation by TTLL8 is not new. This was established back in 2009 when Rogowski et al (2009) showed that polyglycylation of tubulin by TTLL10 occurs only when co-expressed in cells with TTLL3 or TTLL8. So, this part of the study adds very little new information to what was known. "

Our study provides the first in vitro evidence with purified recombinant components that human TTLL8 is exclusively a monoglycylase (Figure 1) and that polyglycylation by TTLL10 requires previous priming with monoglycylation (Figure 2). Studies with purified

recombinant components are the gold standard for establishing the activity of an enzyme as cellular work can be obfuscated by the activity of other regulators. We did cite in our original submission the work by Rogowski, Gaertig and Janke from 2009 (reference 15 in the original submission) as well as that Ikegami and Setou 2009 work (reference 26 in the original submission) that established that TTLL10 polyglycylase activity requires co-expression with TTLL8 in cells. Specifically, we stated in our original submission and in the revised manuscript:

“Cellular overexpression studies coupled with the use of antibodies that recognize mono- and polyglycylation indicate that TTLL8 is also a glycylation-initiator, while TTLL10 a glycylation-elongase (15, 26). However, direct biochemical evidence with purified enzymes for segregated initiation and elongation activity for glycylation is still lacking as does knowledge of their substrate specificity and regulation.”

In addition to citing the Setou study, we now cite again the Rogowski, Gaertig and Janke 2009 study later in the manuscript when the cellular data are mentioned again. Specifically, we state in the revised manuscript:

“This is consistent with cellular overexpression data which showed that polyglycylation signal was detected via antibody only in tubulin from cells that co-expressed TTLL8 and TTLL10, but not TTLL10 alone (15, 26).”

“The study also fails to discuss the involvement of the other monoglycylase, TTLL3 in the entire study, which is a weakness as in vivo, in cells, both the monoglycylases act in concert and so, may play a role in regulating the activity of TTLL10. ”

We previously showed that purified recombinant TTLL3, like TTLL8, adds only monoglycines, with a preference for the b-tubulin tail (Garnham et al., PNAS 2017). Given that TTLL10 requires priming by monoglycylation, we expect that, similarly to TTLL8, TTLL3 will allow elongation of the initial monoglycyl chains by TTLL10.

*(1) From the mass spec data, it appears that the *Xenopus laevis* TTLL10 can add up to 18 residues. However, the numbers indicated in Figure 2E seem to suggest that it is a maximum of 23 residues only at a particular position. Does this mean that the 13-18 residues observed are a collection of multiple short-chain polyglycylation or are there positions that the authors observed where there were chains of longer than 3 glycine residues? This would be an interesting point to note as when it was discovered in *Paramecium*, the polyglycyl chains were reported to be up to 34 residues (Redeker et al., Science 1994). If the authors could test the TTLL10 from *Paramecium* to observe if this is a consistent phenomenon across evolution or is there a biologically significant difference that is being developed, would be interesting to know.”*

Figure 2E shows a subset of the modified tails that we identified and where the position of the posttranslationally added glycine can be mapped to a specific position, or range of positions. Additional species exist. We note that the mass spectra in Figure 2B are intact LC/MS, while those in Figure 2E are MS/MS. The ionization of tubulin tail peptides with larger number of glycines is not as efficient as for shorter glycine chains, reducing the sensitivity of detection of species that have higher number of glycines. This is not as pronounced when the mass spectra are obtained from the intact protein (Figure 2B). In summary, our data supports the fact that TTLL10 elongates polyglycine chains at multiple positions in the tubulin tail (shown in Figure 2E), however, we cannot ascertain the maximum polyglycine chain length, only the total number of glycines added.

Testing the enzyme from *Paramecium* is an interesting proposal but outside the scope of this manuscript.

(2) While it is interesting to know that the TTL10 binds to TTL8-modified tubulin with a much higher affinity than unmodified tubulin, in vivo, the microtubules will be a mixture of both TTL3- and TTL8-modified tubulin. It would be good to see the binding of the enzyme to a tubulin that is modified by both TTL3 and TTL8 if the two have a greater influence on TTL10 binding."

Our previous work showed that purified recombinant TTL3 has purely monoglycylase activity, with a preference for b-tubulin (Garnham et al., PNAS 2017). The sites of monoglycylation by TTL3 overlap with those introduced by TTL8 on b-tubulin (the difference being mainly that TTL3 is more selective towards b-tubulin and thus has lower activity on a-tubulin). TTL8 introduces additional monoGlys on the a-tubulin tail. Therefore, it is unlikely that TTL10 will have a different response to microtubules that carry similar numbers of Gly residues, regardless of whether introduced by TTL8 or TTL3 and 8. Our data show that TTL10 binding increases with Gly number, but that the gains in affinity plateau as the density of glycine residues on the tails increases above a certain threshold, likely because one TTL10 molecule recognizes one monoGly branch, and steric hindrance on the tubulin tail prevents further recruitment of additional TTL10 molecules.

(3) The authors have always increased the number of monoglycines in beta-tubulin more than in alpha-tubulin. Is there a rationale for this? Since TTL8 is known to predominantly modify alphasubulin (Rogowski et al., 2009; Gadadhar et al., 2017) why did the authors not check for the increased binding of the TTL10 on dimers where the number of monoglycines on alpha-tubulin is higher than 1.1? Especially when they themselves observe in their mass spec that even on alphasubulin there are 1, 2, and 3 glycines added. I would like to see what happens if the ratio is high alpha-G + low beta-G"

As our spectra in Figure 1 show, we find that TTL8 is able to modify robustly in vitro both a- and b-tubulin but that it shows a slight preference for b-tubulin (Figure 1B). The work from the Janke group that the reviewer is referring to (Rogowski et al., 2009 and Gadadhar et al., 2017) did not use recombinant, purified enzymes and unmodified microtubules as substrates and used axonemal tubulin (which carries many modifications), and so it is possible that the a-tubulin preference observed in that system when TTL8 is overexpressed, is likely to other factors that do not reflect the biochemical property of the enzyme alone (for example, it could be because btubulin site are not available because they are already glutamylated). As can be seen from Figure 3D, the gain in affinity when increasing the number of glycines from one glycine is small, compared to the initial monoglycine added to the a- and the b-tubulin tail, likely reflecting that one tail cannot bind more than one TTL10 at one time because of steric hindrance. Moreover, it is important here to note that glutamylation and glycylation compete for the same sites on the tubulin tails, as we have for example shown for TTL3 and TTL7 (Garnham et al., 2017), therefore the activity of these enzymes in vivo or with non-naïve substrates are context dependent and influences also what sites are available for TTL10 to modify. In conclusion, by using recombinant enzymes and naïve tubulin we gain insight into the intrinsic property of these enzymes and therefore provide a framework for the interpretation of in vitro and in vivo observations.

(4) I wonder why the authors did not use the human TTL10 to test if this also shows similar binding to the glycylation tubulin despite the fact that it is enzymatically inactive. If it does, then it would be interesting to see the kinetics of binding of this enzyme to see if the fall off of the enzyme from the tubulin is solely driven by the level of polyglycylation only, or if it has any other mechanism involved as well."

Work with human recombinant TTLL10, a TTLL10 homolog which was proposed to be inactive, will be an interesting future direction but outside the scope of this manuscript. We did note in our previous manuscript (Garnham et al., 2017, Figure S5) that the residues which are mutated in the human enzyme compared to other mammals are on the dorsal face of the enzyme, far away from the active site, raising an interesting question of how they inactivate the enzyme. We need however to emphasize that our work clearly shows that it is polyglycylation on the microtubules that reduces binding of TTLL10 to microtubules because experiments done in the absence of glycylation activity i.e. with enzyme that was incubated with microtubules that were pre-modified with polyglycine chains, but in the absence of glycine substrate (precluding any glycylation activity during the binding assay) show that the binding decreases monotonically with the number of polyglycines on the microtubule (Figures 4A, B).

(5) In Figure 5, the authors use monoglycylated tubulin that is either glutamylated or not to show that the activity of TTLL10 is enhanced by the extent of polyglutamylated present on the tubulin. However, there is no evidence of the enzyme binding to microtubules that are only glutamylated. It would be good to test this to determine if the binding is also dependent on both monoglycylation and glutamylated or is it only the enzyme activity.

Figure 5E shows that TTLL10 binding increases with monoglycylation alone, and that glutamylated is additive and Figures 4A, B show that it is not the enzyme activity that affects the binding, but the glycylation state of the microtubule. We did not determine binding to microtubules that were only glutamylated, because TTLL10 would not be able to elongate polyglycine chains on those microtubules, even if it bound.

(6) The level of polyglycylation used in Figure 5 is quite low. It would be good to see how the length of the polyglycine chain impacts TTLL10 activity in the presence of polyglutamylated, and whether this has any cooperative effect leading to longer chain polyglycylation than what is seen with only monoglycylated tubulin.

We expect longer chain polyglycylation to have an inhibitory effect as we show in Figure 4.

“(7) In the overall study, the authors fail to discuss whether the activity of both the glycylation at different sites on tubulin is sequential, or modifications at different residues happen all at once. If the authors were to do a sequential time course of the modification followed by MS/MS analysis, they could get some indications about this.”

As the data in Figure 3D shows, the effect of adding more monoGly site on a tubulin tail has a muted effect on binding, indicating that the additional mono-Gly branches do not lead to more TTLL10 recruitment because of steric hindrance i.e. multiple TTLL10 enzymes cannot be accommodated on the same tail at the same time efficiently. This is consistent with the overall dimensions of the enzyme and the positions of its active site, which were modeled initially in our previous publication (Garnham et al., PNAS 2017). The site of TTLL10 action is pre-determined by the position of the mono-Gly branch introduced by TTLL3 or TTLL8. The length of the tubulin tail and the proximity of mono-Gly sites to each other precludes TTLL10 acting at multiple positions at once on the same tail.

“(8) Do the modifications have any cooperative effect with respect to the sites of modification? Does modifying a particular site enhance the kinetics of modification of the other sites? Can the authors test this?”

This would be an interesting line of future investigations.

"Minor points:

(1) The authors opine that the level of polyglycylation is regulated by the decreased binding of the TTLL10 to the polyglycylation tubulin. While this is an interesting argument, which could be a possibility based on the data they present, it would still not answer if this is a mechanism followed by TTLL10 of all species or not. If they could test the efficacy of TTLL10 from another species, to see the binding efficiency of that enzyme, it could potentially strengthen their argument of this possible mechanism."

The differences between the properties of TTLL10 from different organisms will be an interesting focus of future investigations, but outside the scope of this present study. However, we would like to point out that the level of sequence conservation between TTLL10 makes it unlikely that other TTLL10 do not follow a similar mechanism, albeit with possible differences in the extent of the response. We also note that we have shown that polyglycylation also inhibits binding to the microtubule of the severing enzyme katanin (Szczena et al., Dev. Cell 2022). Therefore, these studies suggests that polyglycylation might be a more general mechanism for reducing microtubule binding affinity since glycylation reduces the negative charge on the tubulin tails, which frequently interact with positively charged domains or interfaces in microtubule associated proteins.

"(2) The authors indicate that glycylation act on pre-glutamylated microtubules. However, in their assays, they use unmodified tubulin, which I would presume is also not glutamylated. If this is the case, how can they justify that the enzymes prefer pre-glutamylated microtubules? This is a bit unclear. Do they mean that their tubulin is already pre-glutamylated? Have they tested this?"

The statement regarding the action of these enzymes on glutamylated microtubules refer to the in vivo situation where polyglycylation microtubules appear in cilia biogenesis after the microtubules in the axoneme are already glutamylated. In vitro, by using microtubules that are only monoglycylation and microtubules that are both glutamylated and monoglycylation, we show that glutamylation further increases recruitment of TTLL10 to microtubules that are monoglycylation. Therefore, glutamylated microtubules will be polyglycylation preferentially over those that are not glutamylated.

We state: "Axonemal microtubules are abundantly glutamylated. Glutamylation appears during cilia development first, followed by glycylation (12, 13), indicating that in this scenario glycylation act on pre-glutamylated microtubule substrates."

"(3) In continuation with the previous point, an immunoblot of their purified tubulin showing no reactivity to anti-glycylation or anti-glutamylation antibodies, which upon treatment with TTLL8 reacts to the anti-glycylation antibody would be confirmatory evidence to show that the isolated tubulin was indeed unmodified."

We have now included a Western blot of our TOG-purified tubulin as Figure S3 in our revised manuscript. This shows a faint signal with the pep-G1 antibody and a very strong signal after TTLL8 treatment. We are not sure whether the low signal with the pep-G1 antibody for the unmodified tubulin is due to low bona fide monoglycylation-specific signal or a low affinity nonspecific interaction of this antibody (raised against mono-glycylation tubulin tail peptides) with the unmodified tubulin. We note that this signal is clearly visible only when loading at least 0.2 micrograms of the purified tubulin. At this loading level the signal for the glycylation species is saturated. It is also important to note that we have not detected glycylation species in this tubulin either by LC-MS or MS/MS. Therefore, our data strongly indicate that the tubulin purified from tsA201 cells is not glycylation or has at most extremely low levels of glycylation. Importantly, this potential trace level of monoglycylation tubulin does not affect

any of the conclusions in this study. The Western blot also shows no detectable signal with the polyglycylation antibody in the unmodified tubulin and a very strong, saturated signal after the tubulin was treated with both TTLL8 and TTLL10. We also added an additional Figure S8 that shows that the tSA201 tubulin does not give a detectable signal for glutamylation. Please see also Figure 3 from Vemu et al., *Methods Enzymology* 2017 where we also published a Western blot from our TOG-purified tubulin using anti-glutamylation antibodies.

"(4) In their study, the authors have used polyglycylation of up to 10-13 residues. This brings me to my first point that in the case of Paramecium, the number was identified to be up to 34, which would mean that this enzyme has higher binding or catalytic activity. I would like to know the authors' perspective on this, as to what could potentially determine the difference in the activities of TTLL10 across species."

The *Xenopus* TTLL10 enzyme can add more glycines than the 10-13 range that we show here if the enzyme is incubated for longer periods. The fact that glycine numbers as high as 34 were detected in *Paramecium* does not necessarily mean that the *Paramecium* enzyme is more active since there is no equivalent data to compare it with from *Xenopus*. The only way to address potential species differences in enzyme specific activity is to purify enzymes from different species and compare their activity side-by-side.

(5) How was the completion of the reaction of monoglycylation and polyglycylation determined? If the enzymes were left for more than 20 minutes, did TTLL8/ TTLL10 add more glycines? What is the reason for using less tubulin (1:20 enzyme:tubulin molar ratio) for monoglycylation by TTLL8, and more tubulin (1:50 enzyme:tubulin molar ratio) for polyglycylation by TTLL10?

Yes, if the enzymes were incubated longer, they added more glycines. The extent of glycylation was determined from the LC-MS and the incubation time was varied to obtain samples with fewer or more glycines. The lower ratio used for TTLL10 is because of the higher specific activity of that enzyme compared to TTLL8.

(6) Figure S2 A, b2 ion is not indicated in the peptide sequence, while it is shown in the m/z graph.

We thank the reviewer for the careful reading. We have corrected this in our MS/MS spectrum.

Reviewer #2 (Public review):

"In their manuscript, Cummings et al. focus on the enzymatic activities of TTLL3, TTLL8, and TTLL10, which catalyze the glycylation of tubulin, a crucial posttranslational modification for cilia maintenance and motility. The experiments are beautifully performed, with meticulous attention to detail and the inclusion of appropriate controls, ensuring the reliability of the findings. The authors utilized in vitro reconstitution to demonstrate that TTLL8 functions exclusively as a glycylation initiator, adding monoglycines at multiple positions on both α - and β -tubulin tails. In contrast, TTLL10 acts solely as a tubulin glycylation elongase, extending existing glycine chains. A notable finding is the differential substrate recognition between TTLL glycylation and TTLL glutamylation, highlighting a broader substrate promiscuity in glycylation compared to the more selective glutamylation. This observation aligns with the greater diversification observed among glutamylation. The study reveals a hierarchical mechanism of enzyme recruitment to microtubules, where TTLL10 binding necessitates prior monoglycylation by TTLL8. This binding is progressively inhibited by increasing polyglycine chain length, suggesting a self-regulatory mechanism for polyglycine chain length control. Furthermore, TTLL10

recruitment is enhanced by TLL6mediated polyglutamylation, illustrating a complex interplay between different tubulin modifications. In addition, they uncover that polyglutamylation stimulates TLL10 recruitment without necessarily increasing glycylation on the same tubulin dimer, due to the potential for TLLs to interact with neighboring tubulin dimers. This mechanism could lead to an enrichment of glycylation on the same microtubule, contributing to the complexity of the tubulin code. The article also addresses a significant challenge in the field: the difficulty of generating microtubules with controlled posttranslational modifications for in vitro studies. By identifying the specific modification sites and the interplay between TLL activities, the authors provide a valuable tool for creating differentially glycylation microtubules. This advancement will facilitate further studies on the effects of glycylation on microtubule-associated proteins and the broader implications of the tubulin code. In summary, this study substantially contributes to our knowledge of posttranslational enzymes and their regulation, offering new insights into the biochemical mechanisms underlying microtubule modifications. The rigorous experimental approach and the novel findings presented make this a pivotal addition to the field of cellular and molecular biology."

We thank the reviewer for their support of our work.

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